

# Generalized Homology and the Eilenberg-Steenrod Axioms

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## Contents

|                                                      |          |
|------------------------------------------------------|----------|
| <b>1 Homotopy Invariance and Homotopy Pushouts</b>   | <b>1</b> |
| 1.1 Graded Functors . . . . .                        | 1        |
| 1.2 Homotopies and Homotopy Invariance . . . . .     | 1        |
| 1.3 Homotopy Pushouts . . . . .                      | 3        |
| <b>2 Homology Axioms and Coefficients</b>            | <b>4</b> |
| 2.1 Homotopy Invariance and Additivity . . . . .     | 4        |
| 2.2 Mayer-Vietoris and Naturality . . . . .          | 4        |
| 2.3 Coefficients and the Dimension Axiom . . . . .   | 6        |
| <b>3 Relative Homology and Spheres</b>               | <b>6</b> |
| 3.1 Reduced Homology and Homotopy Cofibers . . . . . | 6        |
| 3.2 The Long Exact Sequence of a Pair . . . . .      | 7        |
| 3.3 Suspension and the Homology of Spheres . . . . . | 7        |

## 1 Homotopy Invariance and Homotopy Pushouts

The first part of a homology theory is functorial data. The gluing axiom needs a construction that remembers the homotopy along which two maps agree.

### 1.1 Graded Functors

For a space  $W$ , write  $\text{pr}_W : W \rightarrow *$  for its unique map to a point.

Write  $\text{Top}$  for the category of topological spaces and continuous maps, and  $\text{Ab}$  for the category of abelian groups. Let  $\text{Top}_{\text{CW}}$  be the full subcategory of spaces homotopy equivalent to CW complexes. All spaces and subspaces in this lecture are assumed to belong to this subcategory. Disjoint unions and the double mapping cylinders below remain in this subcategory. Homotopy pushouts are understood up to homotopy equivalence. For spaces outside this subcategory, replacing homotopy equivalences by weak equivalences requires an additional convention.

**Definition 1.1.** A graded homology functor is a family of covariant functors

$$A_n : \text{Top}_{\text{CW}} \rightarrow \text{Ab} \quad (n \in \mathbb{Z}).$$

For a map  $f : X \rightarrow Y$ , write  $f_* = A_n(f) : A_n(X) \rightarrow A_n(Y)$ . Functoriality means that identities induce identities and  $(gf)_* = g_*f_*$ . The degree is specified by the source and target when it is omitted from  $f_*$ .

### 1.2 Homotopies and Homotopy Invariance

Write  $I = [0, 1]$ . Given maps

$$f_0, f_1 : X \rightarrow Y,$$

a *homotopy* from  $f_0$  to  $f_1$  is a continuous map

$$\alpha : X \times I \rightarrow Y, \quad \alpha(x, 0) = f_0(x), \quad \alpha(x, 1) = f_1(x).$$

The endpoint inclusions are

$$\iota_0 : X \rightarrow X \times I, \quad x \mapsto (x, 0),$$

$$\iota_1 : X \rightarrow X \times I, \quad x \mapsto (x, 1).$$

Thus the endpoint conditions say that the following triangles commute:

$$\begin{array}{ccccc} X & \xrightarrow{\iota_0} & X \times I & \xleftarrow{\iota_1} & X \\ & \searrow f_0 & \downarrow \alpha & \swarrow f_1 & \\ & & Y & & \end{array}$$

We write  $f_0 \simeq f_1$ . Constant homotopies, reversal of  $I$ , and concatenation show that this is an equivalence relation on the set of maps between two fixed spaces.

**Definition 1.2.** The category  $\mathbf{hTop}_{\text{CW}}$  has spaces as objects and morphism sets

$$\text{Mor}_{\mathbf{hTop}_{\text{CW}}}(X, Y) = \text{Mor}_{\mathbf{Top}_{\text{CW}}}(X, Y) / \simeq.$$

The class of a map  $f$  is written  $[f]$ . For composable maps

$$f : X \rightarrow Y, \quad g : Y \rightarrow Z,$$

composition and identities are defined by

$$[g] \circ [f] = [g \circ f], \quad \text{id}_X = [\text{id}_X].$$

To check composition, take homotopies

$$\alpha : X \times I \rightarrow Y, \quad \beta : Y \times I \rightarrow Z.$$

The composite homotopy is the continuous map

$$\gamma : X \times I \rightarrow Z, \quad (x, t) \mapsto \beta(\alpha(x, t), t).$$

Its endpoints are the composites of the endpoint maps of  $\alpha$  and  $\beta$ . Hence composition of homotopy classes is well defined. Associativity and the identity laws follow from those in  $\mathbf{Top}_{\text{CW}}$ .

The quotient functor is

$$q : \mathbf{Top}_{\text{CW}} \rightarrow \mathbf{hTop}_{\text{CW}}, \quad q(X) = X, \quad q(f) = [f].$$

A *homotopy equivalence* is a map admitting a homotopy inverse:

$$f : X \rightarrow Y, \quad g : Y \rightarrow X, \quad gf \simeq \text{id}_X, \quad fg \simeq \text{id}_Y.$$

Thus  $f$  is a homotopy equivalence exactly when  $[f]$  is an isomorphism in  $\mathbf{hTop}_{\text{CW}}$ , with inverse  $[g]$ .

**Definition 1.3.** Let  $\mathcal{C}$  be a category. A functor

$$F : \text{Top}_{\text{CW}} \rightarrow \mathcal{C}$$

is *homotopy invariant* if homotopic maps have equal images: for

$$f_0, f_1 : X \rightarrow Y, \quad f_0 \simeq f_1,$$

one has

$$F(f_0) = F(f_1) : F(X) \rightarrow F(Y).$$

Equivalently, there is a unique functor

$$\bar{F} : \text{hTop}_{\text{CW}} \rightarrow \mathcal{C}, \quad \bar{F}(X) = F(X), \quad \bar{F}([f]) = F(f),$$

making the triangle commute:

$$\begin{array}{ccc} \text{Top}_{\text{CW}} & \xrightarrow{q} & \text{hTop}_{\text{CW}} \\ & \searrow F & \downarrow \bar{F} \\ & & \mathcal{C}. \end{array}$$

Homotopy invariance is exactly what makes the displayed formula on morphisms well defined. The identity and composition laws descend from  $F$ , and the formulas force uniqueness.

**Proposition 1.4.** Every homotopy invariant functor sends homotopy equivalences to isomorphisms.

**Proof.** For a homotopy equivalence and its homotopy inverse, write

$$f : X \rightarrow Y, \quad g : Y \rightarrow X.$$

Their images have types

$$F(f) : F(X) \rightarrow F(Y), \quad F(g) : F(Y) \rightarrow F(X).$$

Functoriality and homotopy invariance give

$$F(g)F(f) = F(gf) = \text{id}_{F(X)}, \quad F(f)F(g) = F(fg) = \text{id}_{F(Y)}.$$

Thus  $F(g)$  is the inverse of  $F(f)$ . □

### 1.3 Homotopy Pushouts

The gluing axiom uses the double mapping cylinder to retain the homotopy between the two composites.

**Definition 1.5.** For a span  $X \xleftarrow{f} W \xrightarrow{g} Y$ , its double mapping cylinder is

$$M(f, g) = (X + (W \times I) + Y) / \sim,$$

where  $(w, 0) \sim f(w)$  and  $(w, 1) \sim g(w)$ . Here  $+$  denotes a finite disjoint union. Let  $\iota_X : X \rightarrow M(f, g)$  and  $\iota_Y : Y \rightarrow M(f, g)$  be the canonical maps. The cylinder supplies a specified homotopy from  $\iota_X f$  to  $\iota_Y g$ .

A map from  $M(f, g)$  to a space  $Z$  is equivalent to maps  $h : X \rightarrow Z$ ,  $k : Y \rightarrow Z$ , and a homotopy from  $hf$  to  $kg$ . This follows from the quotient universal property: the homotopy defines the map on  $W \times I$ , and its endpoint conditions are precisely the identifications in the quotient.

**Definition 1.6.** Consider maps and a specified homotopy  $\alpha : hf \simeq kg$  in a square

$$\begin{array}{ccc} W & \xrightarrow{f} & X \\ g \downarrow & & \downarrow h \\ Y & \xrightarrow{k} & Z. \end{array}$$

The square, with  $\alpha$ , is a homotopy pushout if its comparison map

$$\theta_\alpha : M(f, g) \rightarrow Z$$

is a homotopy equivalence. This comparison restricts to  $h, k$  and sends  $(w, t)$  to  $\alpha(w, t)$ .

**Remark 1.7.** A commutative pushout square need not be a homotopy pushout. For the span  $* \xleftarrow{\text{pr}_{S^0}} S^0 \xrightarrow{\text{pr}_{S^0}} *$ , where  $S^0$  has two points, the ordinary pushout is a point. Its double mapping cylinder consists of two intervals joining the two target points, hence is a circle. Thus the comparison to the ordinary pushout is not a homotopy equivalence. The specified homotopy in Definition 1.6 is part of the gluing data.

## 2 Homology Axioms and Coefficients

We now impose homotopy invariance, additivity, and Mayer-Vietoris. The connecting homomorphisms in the third axiom are additional structure; their existence is not a consequence of functoriality alone.

### 2.1 Homotopy Invariance and Additivity

Homotopic maps induce the same homomorphism, and disjoint unions correspond to direct sums.

**Axiom 2.1.** Homotopy invariance: for every  $n \in \mathbb{Z}$  and every pair of homotopic maps  $f_0, f_1 : X \rightarrow Y$ ,

$$f_{0,*} = f_{1,*} : A_n(X) \rightarrow A_n(Y).$$

**Axiom 2.2.** Additivity: for every set-indexed family  $(X_i)_{i \in J}$ , the inclusions  $\iota_i : X_i \rightarrow \coprod_{i \in J} X_i$  induce an isomorphism

$$\bigoplus_{i \in J} A_n(X_i) \xrightarrow{\cong} A_n\left(\coprod_{i \in J} X_i\right).$$

The empty family is included, so  $A_n(\emptyset) = 0$ .

### 2.2 Mayer-Vietoris and Naturality

The homotopy in a double mapping cylinder makes  $\iota_{X,*} f_* = \iota_{Y,*} g_*$ . Consequently the composite of the following two maps is zero:

$$A_n(W) \xrightarrow{(f_*, -g_*)} A_n(X) \oplus A_n(Y) \xrightarrow{\iota_{X,*} + \iota_{Y,*}} A_n(M(f, g)).$$

The gluing axiom asserts exactness and supplies the connecting maps that continue this sequence.

**Axiom 2.3.** Mayer-Vietoris: for every span  $X \xleftarrow{f} W \xrightarrow{g} Y$ , there are homomorphisms

$$\delta_n : A_n(M(f, g)) \rightarrow A_{n-1}(W)$$

for which the following sequence is exact in every degree:

$$\begin{aligned} \cdots \rightarrow A_n(W) &\xrightarrow{(f_*, -g_*)} A_n(X) \oplus A_n(Y) \xrightarrow{i_{X,*} + i_{Y,*}} A_n(M(f, g)) \\ &\xrightarrow{\delta_n} A_{n-1}(W) \xrightarrow{(f_*, -g_*)} A_{n-1}(X) \oplus A_{n-1}(Y) \rightarrow \cdots \end{aligned} \quad (1)$$

The homomorphisms  $\delta_n$  are natural for maps of spans and their induced maps of double mapping cylinders.

A map of spans consists of maps

$$\alpha : W \rightarrow W', \quad \beta : X \rightarrow X', \quad \gamma : Y \rightarrow Y'$$

satisfying  $\beta f = f' \alpha$  and  $\gamma g = g' \alpha$ . It induces

$$\mu : M(f, g) \rightarrow M(f', g'),$$

whose restrictions are  $\beta$  on  $X$ ,  $\gamma$  on  $Y$ , and  $(w, t) \mapsto (\alpha(w), t)$  on the cylinder. Naturality of the connecting maps means that

$$\begin{array}{ccc} A_n(M(f, g)) & \xrightarrow{\delta_n} & A_{n-1}(W) \\ \mu_* \downarrow & & \downarrow \alpha_* \\ A_n(M(f', g')) & \xrightarrow{\delta'_n} & A_{n-1}(W') \end{array}$$

commutes. Exactness in (1) means that the image of each homomorphism is the kernel of the next one.

**Definition 2.4.** A generalized homology theory in this formulation consists of the functors of Definition 1.1 and the natural connecting homomorphisms of Axiom 2.3, satisfying Axioms 2.1–2.3. A morphism of theories is a family of natural transformations commuting with the connecting homomorphisms.

**Proposition 2.5.** For a homotopy pushout square as in Definition 1.6, there is a long exact sequence

$$\cdots \rightarrow A_n(W) \xrightarrow{(f_*, -g_*)} A_n(X) \oplus A_n(Y) \xrightarrow{h_* + k_*} A_n(Z) \xrightarrow{\delta_n^Z} A_{n-1}(W) \rightarrow \cdots$$

Here  $\delta_n^Z = \delta_n(\theta_{\alpha,*})^{-1}$ .

**Proof.** Axiom 2.1 sends homotopy equivalences to isomorphisms: the image of a homotopy inverse is an inverse. Thus  $\theta_{\alpha,*}$  is invertible. Transport (1) through this isomorphism. The identities  $\theta_{\alpha} \iota_X = h$  and  $\theta_{\alpha} \iota_Y = k$  identify its middle map with  $h_* + k_*$ . Naturality holds for compatible maps of these squares and their chosen cylinder comparisons.  $\square$

**Remark 2.6.** An open cover  $X = U \cup V$  gives the familiar square

$$\begin{array}{ccc} U \cap V & \hookrightarrow & U \\ \downarrow & & \downarrow \\ V & \hookrightarrow & X. \end{array}$$

For a numerable cover, this is a homotopy pushout. Here numerable means that there is a subordinate partition of unity; for two sets, continuous functions  $\lambda_U, \lambda_V : X \rightarrow [0, 1]$  have sum

1 and supports contained in  $U, V$ , respectively. Such functions give a section of the cylinder comparison by using height  $\lambda_V(x)$  on the overlap. Moving each cylinder coordinate linearly to this height gives a homotopy from the identity to the section followed by the comparison. Thus Proposition 2.5 applies. Open covers of paracompact Hausdorff spaces are numerable. In particular, the covers of spheres used below satisfy this condition.

## 2.3 Coefficients and the Dimension Axiom

The values on a point remain part of the information in a generalized theory. They determine its values on discrete and contractible spaces, but need not be concentrated in degree zero.

**Definition 2.7.** The coefficient groups of  $A$  are the graded abelian groups  $G_n = A_n(*)$ . The dimension axiom is the additional condition  $G_n = 0$  for  $n \neq 0$ . For ordinary integral homology one further chooses  $G_0 \cong \mathbb{Z}$ .

**Proposition 2.8.** For every nonempty contractible space  $X$  and every discrete set  $E$ ,

$$A_n(X) \cong G_n, \quad A_n(E) \cong \bigoplus_{e \in E} G_n.$$

The first isomorphism is induced by the unique map to a point, and the second by the point inclusions.

**Proof.** The map  $\text{pr}_X : X \rightarrow *$  is a homotopy equivalence, so Axiom 2.1 gives the first isomorphism. A discrete set is a disjoint union of points; Axiom 2.2 gives the second, including  $E = \emptyset$ .  $\square$

**Example 2.9.** Once ordinary integral homology satisfies the axioms, its shift  $A_n(X) = H_{n-1}(X; \mathbb{Z})$  also satisfies them: shift every degree in its Mayer-Vietoris sequence and connecting maps. Its only nonzero coefficient group is  $G_1 = \mathbb{Z}$ . This theory violates the dimension axiom while retaining homotopy invariance, additivity, and Mayer-Vietoris. A more general family is  $A_n(X) = \bigoplus_{r \in \mathbb{Z}} H_{n-r}(X; B_r)$  for a fixed graded family of coefficient groups  $(B_r)$ . Direct sums of exact sequences are exact, so the same argument applies.

**Remark 2.10.** Generalized theories have groups in every integer degree; their long exact sequences do not stop at degree zero. Ordinary singular homology is obtained with its dimension normalization and zero negative-degree groups. The Eilenberg-Steenrod uniqueness theorem identifies ordinary theories with fixed coefficients on CW complexes, up to an isomorphism respecting that normalization. The proof uses cellular induction together with an argument for infinite complexes. See May, Chapter 15, Section 2.

## 3 Relative Homology and Spheres

The homotopy cofiber defines relative homology. Its Mayer-Vietoris sequence yields the long exact sequence of a pair.

### 3.1 Reduced Homology and Homotopy Cofibers

Write  $*$  for a one-point space. Choosing a point of a space splits off its coefficient summand.

**Definition 3.1.** For a space  $Y$  with a chosen point  $y_0$ , let  $\text{pr}_Y : Y \rightarrow *$  be the unique map and  $s : * \rightarrow Y$  select  $y_0$ . Define

$$\tilde{A}_n(Y) = \ker((\text{pr}_Y)_* : A_n(Y) \rightarrow G_n).$$

Since  $(\text{pr}_Y)_* s_* = \text{id}$ , there is an isomorphism  $A_n(Y) \cong \tilde{A}_n(Y) \oplus G_n$ . Explicitly,  $x$  is sent to  $(x - s_*(\text{pr}_Y)_* x, (\text{pr}_Y)_* x)$ .

**Definition 3.2.** For a subspace inclusion  $\iota_X : X' \hookrightarrow X$ , define its homotopy cofiber  $C(\iota_X)$  to be the double mapping cylinder of  $X \xleftarrow{\iota_X} X' \xrightarrow{\text{pr}_{X'}} *$ . Its distinguished point is the copy of  $*$ . Define

$$A_n(X, X') = \tilde{A}_n(C(\iota_X)).$$

A map of pairs is a commutative square

$$\begin{array}{ccc} X' & \xhookrightarrow{\iota_X} & X \\ f' \downarrow & & \downarrow f \\ Y' & \xhookrightarrow{\iota_Y} & Y. \end{array}$$

It gives a basepoint-preserving map  $C(\iota_X) \rightarrow C(\iota_Y)$ , hence a homomorphism of relative groups. Functoriality follows directly from the cylinder construction.

### 3.2 The Long Exact Sequence of a Pair

Mayer-Vietoris for a homotopy cofiber gives the relative exact sequence.

**Proposition 3.3.** Every pair has a natural long exact sequence

$$\cdots \rightarrow A_n(X') \xrightarrow{\iota_{X,*}} A_n(X) \xrightarrow{q_*} A_n(X, X') \xrightarrow{\partial_n} A_{n-1}(X') \rightarrow \cdots.$$

**Proof.** Let  $\kappa : X \rightarrow C(\iota_X)$  and  $s : * \rightarrow C(\iota_X)$  be the two canonical maps. Mayer-Vietoris gives

$$\cdots \rightarrow A_n(X') \xrightarrow{(\iota_{X,*}, -(\text{pr}_{X'})_*)} A_n(X) \oplus G_n \xrightarrow{\kappa_* + s_*} A_n(C(\iota_X)) \xrightarrow{\delta_n} A_{n-1}(X') \rightarrow \cdots,$$

where  $\text{pr}_{X'} : X' \rightarrow *$  is the unique map. Under the splitting of Definition 3.1, set  $q_*x = \kappa_*x - s_*(\text{pr}_X)_*x$  and let  $\partial_n$  be the restriction of  $\delta_n$  to the reduced summand. Change coordinates on the middle direct sum by  $(x, g) \mapsto (x, (\text{pr}_X)_*x + g)$ . The first map becomes  $a \mapsto (\iota_{X,*}a, 0)$ ; the next becomes  $(x, g) \mapsto (q_*x, g)$ . Also  $\delta_n s_* = 0$  by exactness. Thus the displayed sequence splits into the claimed sequence and the identity map  $G_n \rightarrow G_n$ . Exactness and naturality pass to the claimed summand.  $\square$

**Remark 3.4.** For a CW pair, or a subspace inclusion satisfying the homotopy extension property, the map from the mapping cone to  $X/X'$  is a homotopy equivalence (with a disjoint basepoint when  $X' = \emptyset$ ). Thus the relative groups are the reduced groups of this quotient. For an arbitrary subspace, the homotopy cofiber is the construction to use; the ordinary quotient need not have its homotopy type. The homotopy extension property says that a homotopy on  $X'$  extending a map on  $X$  extends to a homotopy on  $X$ . See May, Chapter 6 and Chapter 14, Section 2.

### 3.3 Suspension and the Homology of Spheres

The homology of spheres is determined by the graded coefficient groups  $G_n$ .

**Proposition 3.5.** For every  $m \geq 0$  and every  $n \in \mathbb{Z}$ ,

$$\tilde{A}_n(S^m) \cong G_{n-m}, \quad A_n(S^m) \cong G_n \oplus G_{n-m}.$$

The reduced group is defined using a chosen point of the sphere.

**Proof.** For  $S^0 = \{x_0, x_1\}$ , additivity identifies  $A_n(S^0)$  with  $G_n \oplus G_n$ . The map to a point is addition, whose kernel consists of  $(x, -x)$  and is isomorphic to  $G_n$ .

For  $m \geq 1$ , cover  $S^m$  by the complements  $U, V$  of its two poles. Both are contractible, and their intersection deformation retracts onto  $S^{m-1}$ . The resulting square is a homotopy pushout by Remark 2.6. Under the coefficient identifications, its first map is

$$A_n(S^{m-1}) \rightarrow G_n \oplus G_n, \quad x \mapsto ((\text{pr}_{S^{m-1}})_*x, -(\text{pr}_{S^{m-1}})_*x).$$

Its image is the antidiagonal copy of  $G_n$ ; its kernel is  $\tilde{A}_n(S^{m-1})$ . Exactness gives

$$0 \rightarrow G_n \rightarrow A_n(S^m) \xrightarrow{\delta_n} \tilde{A}_{n-1}(S^{m-1}) \rightarrow 0.$$

The first homomorphism is the map induced by a chosen point in  $U$ , so the map  $\text{pr}_{S^m} : S^m \rightarrow *$  splits it. Hence  $\delta_n$  restricts to an isomorphism from  $\tilde{A}_n(S^m)$  onto  $\tilde{A}_{n-1}(S^{m-1})$ . Induction proves the formula.  $\square$

**Example 3.6.** With  $G_0 = \mathbb{Z}$  and  $G_n = 0$  for  $n \neq 0$ , Proposition 3.5 gives the usual integral homology of spheres. For the shifted theory of Example 2.9, the additional sphere summand occurs in degree  $m + 1$ , and the coefficient summand occurs in degree 1. A connected space can therefore have several nonzero groups unrelated to its dimension.

## Reference

J. P. May, *A Concise Course in Algebraic Topology*, Chapters 6, 13–15: mapping cylinders, homology axioms, reduced and relative homology, Mayer-Vietoris, and uniqueness on CW complexes. The gluing-first formulation above uses homotopy pushouts in place of a separate excision axiom.

<https://www.math.uchicago.edu/~may/CONCISE/ConciseRevised.pdf>